

Berke Göktuğ Sarılı

Automation & SCADA Engineer | Electrical & Electronics Engineer

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PROFESSIONAL SUMMARY

Automation engineer with end-to-end experience in industrial automation and digitalization: Siemens PLC programming (S7-300/400/1200/1500), SCADA development (WinCC RT Professional), industrial data infrastructure (Kepware KEPServerEX OPC server, Microsoft SQL Server) and electrical design (EPLAN). Delivered projects in waste-to-energy, recycling, water treatment, hydraulic press machinery and CNC systems — from cabinet design and PLC software to commissioning, retrofit and troubleshooting. Also designs and ships production-grade web applications (React, TypeScript, Node.js) that digitalize field operations, including a live route-optimization and fleet-tracking platform (rotadoa.com). Active in Industry 4.0 initiatives: predictive maintenance, machine-learning-based fault prediction and industrial IoT.

CORE SKILLS

PLC & Motion Control	Siemens TIA Portal (S7-300/400/1200/1500), safety logic, Siemens LOGO!, Pilz PLC, servo & drive commissioning, Sinumerik ONE CNC programming
SCADA & HMI	Siemens WinCC RT Professional, HMI design, alarm management, trending & automated reporting
Industrial Data & Comm.	Kepware KEPServerEX, OPC UA/DA, Microsoft SQL Server, PROFINET (wired & wireless), Modbus RTU/TCP, LoRaWAN, Siemens SCALANCE routers, industrial cybersecurity
Software Development	Python, TypeScript/Javascript, React, Node.js (Express), REST API, WebSocket, SQL, Prisma ORM, Git, PWA development
Electrical Design	EPLAN Electric P8, Proteus, Eagle (PCB); high-voltage operation responsibility (EMO certified)
Embedded Systems	STM32, Arduino, C/C++
Languages	Turkish (native), English (C1)

PROFESSIONAL EXPERIENCE

Automation Engineer — ITC Invest Trading & Consulting AG

Ankara, Türkiye | February 2026 – Present | Recycling & waste management group: waste-to-energy incineration, medical waste sterilization, water treatment facilities

- Responsible for the end-to-end automation infrastructure of multiple plants: Siemens S7-1500 PLC programming, WinCC RT Professional SCADA development, alarm management, trending and automated reporting; provide real-time troubleshooting for production-line and industrial system failures across facilities.
- Re-engineered a medical waste sterilization machine end to end: wrote the complete control software from scratch (TIA Portal, WinCC RT Professional), implemented the sterilization cycle logic with automated PDF cycle-report generation, and integrated the machine into the plant automation system.
- Designed, developed and launched **rotadoa.com** — a route optimization and live fleet-tracking platform for the company's collection vehicles: traffic-aware routing with alternative route selection (TomTom / OpenRouteService / OSRM fallback chain), real-time multi-vehicle map tracking over WebSocket, automated SMS notifications with delivery-code verification, and vehicle fill-rate analytics. Stack: React + TypeScript PWA, Node.js (Express), Prisma ORM, JWT role-based access control (admin / manager / driver).
- Developed a **facility inspection application** that replaced paper-based plant control forms: KKS-coded equipment checklists, photo documentation and importance-weighted scoring; deployed as a PWA over HTTPS on the company Linux server and used on tablets in the field.
- Built a **true-to-scale (mm) electrical panel layout tool** for panel assembly teams: drag-and-drop component placement, automatic layout suggestions and printable PDF outputs; deployed on the company server.
- Commissioned a **crane automation system with wireless PROFINET**: established antenna-based wireless PROFINET communication between the moving structure and the control system, ensuring reliable, continuous communication in an application where cabling was not feasible (TIA Portal).

- Established PLC–SCADA–SQL data pipelines with KEPServerEX and SQL Server, delivering dashboards and periodic reports that support operational decision-making and performance analysis.
- Root-caused chronic SCADA client performance problems across three plants (network adapter / DNS misconfiguration) and rolled out a standardized network configuration guideline.
- Carried out Industry 4.0 studies using image processing and machine learning, including gearbox failure-time estimation and gas-sensor fault-prediction models for predictive maintenance.
- Maintain GSM-based industrial alarm/SMS systems (Siemens SCALANCE M874-3) and remote-site PLC communications; execute PLC CPU migration and retrofit projects in TIA Portal.

Engineering Lieutenant (Military Service) — Turkish Air Force, Akıncı Air Base

Ankara, Türkiye | **November 2024 – November 2025**

- Managed high-voltage infrastructure of the air base: transformer substation maintenance, lightning protection systems and electrical surveys/acceptance procedures for new buildings, in full compliance with military standards and safety regulations.
- Coordinated technical teams during field operations, strengthening leadership and engineering management skills.
- In parallel with official duties, continued independent automation studies: asynchronous motors and drives, servo motor control, analog I/O, industrial sensors, communication protocols, EPLAN design and Python-based control projects.

Automation Engineer — Dirinler Machinery

İzmir, Türkiye | **July 2023 – November 2024** | Industrial hydraulic press machines & CNC systems

- Designed H-type (CDCH/CDHH) and C-type (CDCS) hydraulic press machines from 63 to 1500 tons in EPLAN and programmed them with Siemens S7-300/1200/1500 PLCs.
- Integrated and commissioned servo-driven manipulators, transfer systems, die clamping mechanisms, automatic stroke control, heating panels and safety encoders.
- Implemented machine safety systems — light curtains, emergency stop circuits, safety encoders — within PLC control logic; hands-on experience with hydraulic and pneumatic systems.
- Developed PLC software for 3–5 axis gantry CNC machines and rail-based underground turning systems used in CNC portal milling applications (Sinumerik-based).
- Performed HMI development; prepared quality-control documentation, user manuals and safety reports; diagnosed and resolved CNC machine faults, minimizing production downtime.

R&D Engineer (Defense Industry) — KZ Mechatronics

Balıkesir, Türkiye | **January 2023 – April 2023**

- Designed PCB layouts in Eagle and Proteus, selected integrated circuits and managed the related production processes.
- Developed embedded software on STM32 and Arduino platforms; validated prototypes of security, monitoring and automation devices (motion, magnetic and light sensors) through field and laboratory testing.

Additional Experience

- **Intern — Ordem Engineering, Ankara:** fire alarm, CCTV and access control systems; installation, configuration and field applications.
- **Sales & Customer Relations — Cheries Bike & Blade Rental, Virginia, USA (June – September 2022):** daily rental and sales operations in an English-speaking environment (Work & Travel program).

EDUCATION

B.Sc. in Electrical & Electronics Engineering — Balıkesir University

2019 – 2023 | GPA: 3.39 / 4.00

- Graduation project: LoRaWAN-based long-range (up to 15 km) PLC communication system operating without internet infrastructure — integrated a Siemens S7-1200 PLC via Modbus to transfer remote boiler-room temperature data to a central control system and drive a fire-sprinkler control loop in real time.
- Delivered PLC laboratory training and hands-on programming demonstrations to students.

Erasmus+ Exchange — Politechnika Częstochowska, Poland

2021 – 2022

- Control systems coursework and laboratory applications; computer-engineering software courses; international project and team management activities.

CERTIFICATES & TRAINING

- Operation Responsibility in High-Voltage Facilities — Chamber of Electrical Engineers (EMO), 04/2025
- Industrial Communication Systems & Cybersecurity in Automation — Udemy, 06/2025
- Programmable Logic Controllers (PLC) Training — Ministry of National Education, 05/2025
- Internet of Things (IoT) — Ministry of National Education, 05/2025
- EPLAN Training — Udemy, 09/2025
- Object-Oriented Programming with Python — Gazi University, 08/2025
- Python Programming — Udemy, 10/2025
- Authorized Boiler Operator License — Ministry of National Education, 03/2025
- English Proficiency Certificate (C1) — Erasmus+, 10/2020 · Upper-Intermediate English (C+) — Perfect English, 08/2023
- Entrepreneurship Training — KOSGEB, 05/2022 · Microprocessor Programming — Udemy, 04/2021